Detects noise?
AUROC clean vs 0 dB (same beats)

	CirCor	ZCH	BMD-HS	PASCAL
B1 Springer SQI	0.89	0.94	0.94	0.95
B2 Das RF	0.86	0.97	0.94	0.94
B3 CardioPHON	0.87	0.98	0.98	0.97
B4 CNN	0.92	0.99	1.00	0.98
B5 CRNN	0.86	0.99	1.00	0.95
B6 PCGKolect	0.96	0.88	0.94	0.97
B7 SNR rule	0.83	0.84	0.85	0.88
B8 PCGKolect-v3	0.88	0.69	0.21	0.88
H-AND Das∧B4	0.90	0.99	0.96	0.97
H-AND Das∧B5	0.87	0.99	0.96	0.96
H-CASCADE Das→B4	0.89	0.97	0.95	0.96
H-CASCADE Das→B5	0.85	0.97	0.95	0.94
H-FUSE Das+B4	0.91	0.99	1.00	0.99
H-FUSE Das+B5	0.80	0.99	1.00	0.96

Does the filter score pathology as worse quality?
AUROC pathological vs normal clean beats (0.5 = blind; red = pathology looks less usable)

	CirCor		ZCHSound				BMD-HS†					PASCAL†		Synthetic rhythm	
	Systolic murmur	Diastolic murmur*	ASD	VSD	PFO	PDA	AS	AR	MR	MS	Multi-valve	Murmur	Extrasystole	Premature (synth.)	AF-like (synth.)
B1 Springer SQI	0.39	0.56	0.45	0.54	0.56	0.59	0.52	0.38	0.67	0.50	0.61	0.45	0.48	0.68	0.69
B2 Das RF	0.51	0.50	0.47	0.48	0.53	0.56	0.47	0.56	0.63	0.63	0.59	0.60	0.48	0.59	0.60
B3 CardioPHON	0.44	0.49	0.67	0.60	0.77	0.71	0.42	0.48	0.56	0.58	0.55	0.59	0.51	0.60	0.61
B4 CNN	0.49	0.59	0.67	0.61	0.78	0.64	0.43	0.35	0.63	0.62	0.51	0.58	0.46	0.48	0.46
B5 CRNN	0.54	0.55	0.61	0.63	0.79	0.66	0.52	0.43	0.65	0.55	0.56	0.61	0.45	0.45	0.44
B6 PCGKolect	0.64	0.66	0.62	0.63	0.75	0.69	0.74	0.60	0.76	0.62	0.69	0.45	0.41	0.91	0.75
B7 SNR rule	0.46	0.75	0.54	0.61	0.60	0.56	0.65	0.55	0.59	0.55	0.64	0.54	0.57	0.46	0.45
B8 PCGKolect-v3	0.57	0.75	0.48	0.54	0.51	0.59	0.48	0.45	0.70	0.54	0.60	0.45	0.46	0.61	0.49
H-AND Das∧B4	0.49	0.53	0.59	0.54	0.64	0.63	0.45	0.43	0.67	0.66	0.58	0.57	0.48	0.53	0.52
H-AND Das∧B5	0.52	0.55	0.51	0.54	0.63	0.61	0.50	0.50	0.68	0.62	0.61	0.60	0.46	0.52	0.51
H-CASCADE Das→B4	0.49	0.52	0.56	0.54	0.65	0.64	0.46	0.50	0.64	0.64	0.58	0.59	0.46	0.53	0.52
H-CASCADE Das→B5	0.52	0.54	0.51	0.54	0.64	0.65	0.49	0.53	0.66	0.61	0.60	0.59	0.45	0.52	0.51
H-FUSE Das+B4	0.49	0.57	0.66	0.59	0.81	0.68	0.44	0.38	0.59	0.57	0.52	0.57	0.46	0.50	0.50
H-FUSE Das+B5	0.52	0.56	0.61	0.59	0.78	0.67	0.50	0.48	0.64	0.56	0.58	0.60	0.44	0.48	0.47

Left: every filter responds to noise. Right: a filter with no pathology bias stays near 0.5; red cells explain the biases (e.g. B3/B4/B5 on ZCHSound PFO, B6 on synthetic rhythm). B2 is noise-sensitive but pathology-blind (≈0.5 everywhere). Seeded models and hybrids: mean AUROC of 3 seeds.